

Substrate-Comparison Monitoring of Oyster Spat Settlement

A general, transferable method for ranking settlement substrates and tracking wild oyster recruitment, growth and survival using a compartmented spat collector.

Protocol ID: hossain_2013

Sector: Mollusc · Usability Tier 4 (Practitioner)

MEL indicators: H&B 1.1.1 (Epibiota / spat settlement) · H&B 2.2.1 (Spawning / recruitment)

Source paper: Oyster aquaculture for coastal defense with food production in Bangladesh

Hossain, M.S., et al. (2013). Aquaculture Asia XVIII(1): 15–24, NACA. Access: <https://www.aquaasiapac.com/>

Introduction

This protocol ranks which settlement substrate best recruits wild oyster spat at a site, and tracks how recruitment, growth and survival develop over a settlement season. Candidate substrates are held side by side in the separate compartments of a spat collector placed on the intertidal flat, so they experience identical conditions. The same framed area of each substrate is re-photographed on a fixed schedule, and spat are counted and measured from the images to give a density, a size, and a survival trend for each substrate.

The method is deliberately gear-agnostic: build the collector from whatever durable, locally-available materials you have, and use your preferred water-quality instruments. It suits low-cost, community-led monitoring in sheltered, shallow, often turbid settings.

Measured parameters

- **Spat density** — settled spat per unit area (individuals m⁻²), from counts within a fixed quadrat area.
- **Substrate type** — categorical; the candidate substrates being compared (e.g. windowpane shell, live oyster shell, dead oyster shell, stone/rubble).
- **Spat size / growth** — shell length and width (mm) measured from the photographs. Length = the longest anterior–posterior axis, roughly parallel to the hinge; width = umbo to ventral shell margin.
- **Survival over time** — the same counts repeated across the season, giving recruitment then post-settlement survival per substrate.
- **Water quality (per visit)** — a recommended core of temperature (°C), salinity (‰), pH, dissolved oxygen (mg L⁻¹) and suspended sediment / turbidity; add current speed, tidal magnitude, wave height, primary productivity, or dissolved/suspended solids if relevant to your site.
- **Site extent (optional)** — reef / shoreline footprint from satellite imagery and GIS.

Requirements

Personnel: 1–2 people. *[estimate]*

Estimated time:

- Preparation (build collectors + reference frame): on the order of a day for a small set.
- Deployment: about half a person-day per site.
- Each monitoring visit: roughly 20–40 minutes per collector (photograph every compartment + water quality).
- Duration: one full settlement season (several months); revisit fortnightly, timed to the full and new moon (spring tides) for consistent low-tide access.

Sampling frequency & timing

Framework guidance. The MEL framework sets a recommended sampling frequency for each indicator, on the principle that sampling should be spaced “at intervals sufficient to build an accurate depiction of the process over time.” Its values run Once · Seasonal · Biannual · Annual · Ongoing, often anchored to a point in the season.

What the source did. Sampled fortnightly — timed to the spring tides of the full and new moon — through a single settlement season.

Recommended for sound science. Sample fortnightly across the settlement season so you capture the settlement-then-survival curve rather than a single snapshot. Then repeat over at least two to three settlement seasons *[set to your local recruitment window]* — spat recruitment varies strongly from year to year, so one season cannot separate a real substrate signal from an unusual year.

Materials

- A multi-compartment spat collector that holds each candidate substrate in its own compartment *[build from preferred, durable, locally-available materials — e.g. a framed mesh mat, trays, or a bamboo frame]*.
- Candidate substrates to compare, one per compartment *[choose substrates available in your area]*. Count or weigh the starting quantity of each before loading, so settlement is comparable.
- A fixed reference frame of known area to scale the photos *[record its dimensions; ~38 × 38 cm works well]*.
- A camera (a good smartphone is fine).
- Water-quality instruments *[use preferred gear — e.g. a multiparameter sonde, refractometer, Secchi disk]*.
- A way to anchor the collector into the substrate *[use preferred method — e.g. poles or stakes]*.
- Waterproof tags/labels; clipboard with the datasheet on waterproof paper; pencil; hand-held GPS.
- (Optional) satellite imagery + GIS software for site mapping.

Methods

Preparation

1. Build one or more multi-compartment spat collectors *[preferred materials]*, so the candidate substrates sit side by side under identical conditions.
2. Count or weigh the starting quantity of each substrate and record it, then load one substrate per compartment.
3. Decide how many collectors (replicates) to deploy per site. *[≥3 recommended for a defensible comparison]*
4. Prepare the fixed reference frame that will appear in every photo *[record its size]*.

Deployment

1. Choose a sheltered intertidal / soft-sediment site — ideally where oysters already settle on nearby hard surfaces such as pilings, sluice gates, or concrete.
2. Set the collectors on the flat and anchor them *[preferred method]*; reinforce against waves and currents.
3. Record site, coordinates, date, the substrate-to-compartment layout, starting substrate quantities, and a deployment photo of each collector.

Monitoring (repeat fortnightly, on the full and new moon)

1. At each visit, photograph the same framed area of every compartment straight-on, with the reference frame in view.
2. From the photos, count settled spat in the frame and measure the length and width of the spat. This yields density, size and — across visits — a survival trend per substrate.
3. Record the water-quality parameters and note conditions (recent storms, heavy freshwater input, sediment on the substrate, fouling).
4. Keep to the fixed interval so trajectories are comparable. *[set and record a minimum recognisable spat size so counts stay consistent between visits and counters]*

Analysis

Use the paired stats workbook. Per-substrate mean density, spat size, and settlement-over-time plots are handled in the workbook. To rank overall substrate performance across all visits at once, run a PCA on the spat-abundance × sampling-period matrix and read vector length / separation to rank the substrates; the workbook's *Stats directions* sheet gives an R recipe.

Expect these failure modes (and record them)

- Fouling competition — anemones, snails and barnacles colonise the substrate and cut the space available to spat. Note fouling cover at each visit.
- Sediment smothering — heavy rain / high river discharge can drop several centimetres of sediment on the substrate and kill or delay spat. Note siltation events.
- Collector durability — many materials weaken in seawater and storms; plan for repairs and reinforcement through the season.
- Predation — crabs, shorebirds and starfish are common on intertidal beds and can remove spat (often unquantified).
- Design tip — installing substrates vertically may improve spatfall relative to flat placement.
- Seasonal effect — expect heavy post-settlement mortality where the season brings low salinity and high siltation.

How this serves the MEL framework

- **H&B 1.1.1 — Epibiota / spat settlement:** direct — counts of settled spat on the deployed substrates.
- **H&B 2.2.1 — Spawning / recruitment:** the fortnightly trajectory captures the recruitment signal and its decline over the season.

Data handling

- Label every photo by collector ID, compartment/substrate, and visit date.
- Enter counts, spat sizes and water-quality readings into the paired workbook; keep the labelled photos as the raw record.

Source provenance & residual gaps

Normalized by AquaPIP from Hossain et al. (2013). The source does not fix: collector / compartment dimensions, replicate count, the instrument or method per water-quality parameter, or the photo spat-counting rules (min size, software) — shown in [brackets] above. The detailed photographic count / measure technique is defined by cited method papers (Loosanoff 1966; O'Sullivan 1978; Davenport & Glasspool 1987; Hendriks et al. 2005), not this article.

Note: this protocol omits the CC 3.1.1 (wave attenuation) mapping carried on the database row, because the method does not monitor coastal protection directly. Protocol ID *hossain_2013*.